

presentation of the method. See also the historical notes in Appendix C of M. F. Gardner and J. L. Barnes, *Transients in Linear Systems* (John Wiley & Sons, Inc., New York, 1942).

DIGITAL COMPUTER EXERCISES

In the study of the Laplace transformation, two operations that are routinely accomplished with the aid of the computer are the determination of the roots of a polynomial and the calculation of residues. References relating to these operations are given in Appendices E-1 and E-5.1. In particular, Chapter 8 of reference 7 by Huelsman in Appendix E-10 which relates to the manipulation of polynomials and the determination of residues is recommended. See also Case Study 23 given by McCracken, reference 12 cited in Appendix E-10.

PROBLEMS

- 7-1. Verify that the Laplace transform of $\cos \omega t$ determined from Eq. (7-1) is

$$\mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2}$$

In each of the problems that follow, repeat the procedure of Prob. 7-1 for the various transform pairs of the following table.

	$f(t)$ for $t \geq 0$	$F(s)$
7-2.	t^2	$\frac{2}{s^3}$
7-3.	$\sinh \alpha t$	$\frac{\alpha}{s^2 - \alpha^2}$
7-4.	$\cosh \alpha t$	$\frac{s}{s^2 - \alpha^2}$
7-5.	$e^{-\alpha t} \sin \omega t$	$\frac{\omega}{(s + \alpha)^2 + \omega^2}$
7-6.	$e^{-\alpha t} \cos \omega t$	$\frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$
7-7.	$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
7-8.	$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$

For the following $f(t)$, determine the other member of the transform pair, $F(s) = \mathcal{L}[f(t)] = P(s)/Q(s)$ where $P(s)$ and $Q(s)$ are polynomials in s .

7-9. $f_1(t) = \sin^2 t$

7-10. $f_2(t) = t \cos at$

7-11. $f_3(t) = [1/(2a)] \sin at$

7-12. $f_4(t) = \cos^2 t$

7-13. $f_5(t) = [1/(2a^3)](\sinh at - \sin at)$

7-14. $f_6(t) = [1/(2a^2)] \sin at \sinh at$

7-15. $f_7(t) = [1/(2a^2)](\cosh at - \cos at)$

7-16. $f_8(t) = [1/(2a)](\sin at + at \cos at)$

7-17. $f_9(t) = t^2 e^{-at} \cos \omega t$

7-18. $f_{10}(t) = (1/t) \sin^2 \omega t$

7-19. $f_{11}(t) = t^{1/2}$

7-20. $f_{12}(t) = e^{-at^2}$

- 7-21. In the network shown in the figure, C is charged to V_0 , and the switch K is closed at $t = 0$. Solve for the current $i(t)$ using the Laplace transformation method.

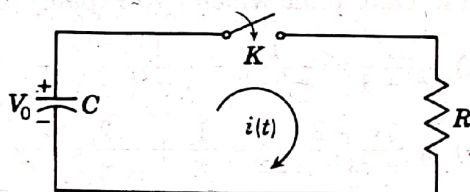


Fig. P7-21.

- 7-22. In the network shown in the figure, the switch K is moved from position a to position b at $t = 0$, a steady state having previously been established at position a . Solve for the current $i(t)$, using the Laplace transformation method.

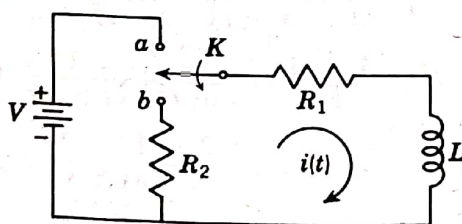


Fig. P7-22.

- 7-23. In the network shown, C is initially charged to V_0 . The switch K is closed at $t = 0$. Solve for the current $i(t)$, using the Laplace transformation method.

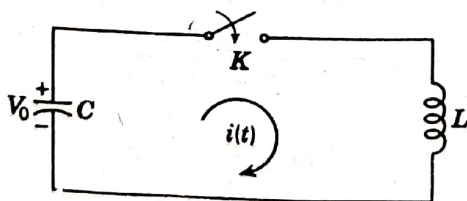


Fig. P7-23.

- 7-24. In the network shown, the switch K is moved from position a to position b at $t = 0$ (a steady state existing in position a before $t = 0$). Solve for the current $i(t)$, using the Laplace transformation method.

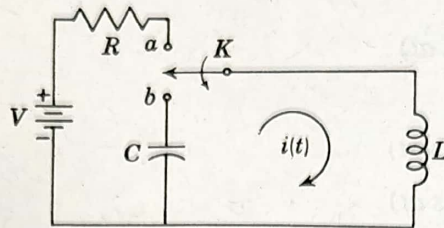


Fig. P7-24.

- 7-25. Work Prob. 4-2 using the Laplace transformation method.
- 7-26. Work Prob. 4-3 using the Laplace transformation method of this chapter.
- 7-27. Work Prob. 4-4 using the Laplace transformation method.
- 7-28. Work Prob. 4-5 using the Laplace transformation method rather than the classical method of Chapter 4.
- 7-29. Check the following equations by completing a partial fraction expansion. Determine which two expansions in the set are in error.
- $\frac{2s}{s^2 - 1} = \frac{1}{s + 1} + \frac{1}{s - 1}$
 - $\frac{7s + 2}{s^3 + 3s^2 + 2s} = \frac{1}{s} + \frac{2}{s + 2} + \frac{-3}{s + 1}$
 - $\frac{5s + 13}{s^2 + 5s + 6} = \frac{2}{s + 3} + \frac{3}{s + 2}$
 - $\frac{s^2}{s - 1} = \frac{1}{s - 1} + s + 1$
 - $\frac{2(s + 1)}{s^2 + 1} = \frac{1 + j1}{s + j1} + \frac{1 - j1}{s - j1}$
 - $\frac{s^2 + 4s + 1}{s(s + 1)^2} = \frac{1}{s + 1} + \frac{1}{(s + 1)^2} + \frac{2}{s}$
 - $\frac{3s^3 - s^2 - 3s + 2}{s^2(s - 1)^2} = \frac{1}{s} + \frac{2}{s^2} + \frac{2}{s - 1} + \frac{1}{(s - 1)^2}$
 - $\frac{s^3 - 5s^2 + 9s + 9}{s^2(s^2 + 9)} = \frac{1}{s} + \frac{1}{s^2} + \frac{-j}{s + j3} + \frac{+j}{s - j3}$
- 7-30. Expand the following functions as partial fractions:

(a) $F_1(s) = \frac{(s + 1)(s + 3)}{s(s + 2)(s + 4)}$

(b) $F_2(s) = \frac{(s^2 + 1)(s^2 + 3)}{s(s^2 + 2)(s^2 + 4)}$

(c) $F_3(s) = \frac{s}{s^2(s + 1)^2(s + 2)}$

$$(d) F_4(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

$$(e) F_5(s) = \frac{s + 4}{(s + 1)^2(s + 3)^2}$$

$$(f) F_6(s) = \frac{s^2(s + 3)}{(s + 1)(s + 2)^4}$$

7-31. Expand the following functions as partial fractions:

$$(a) F_a(s) = \frac{s^2 + 7s + 8}{(s + 1)(s + 2)(s + 3)}$$

$$(b) F_b(s) = \frac{s^2 + s + 1}{(s + 1)(s + 2)(s + 3)}$$

$$(c) F_c(s) = \frac{s + 4}{s(s + 1)(s + 3)^2(s + 2)^3}$$

7-32. Verify the following inverse Laplace transformations $\mathcal{L}^{-1}F(s) = f(t)$:

$$(a) \mathcal{L}^{-1} \frac{3s}{(s^2 + 1)(s^2 + 4)} = \cos t - \cos 2t$$

$$(b) \mathcal{L}^{-1} \frac{s + 1}{s^2 + 2s} = \frac{1}{2}(1 + e^{-2t})$$

$$(c) \mathcal{L}^{-1} \frac{1}{s(s^2 - 2s + 5)} = \frac{1}{5}[1 + \frac{1}{2}e^{-(2t)}(-2 \cos 2t + \sin 2t)]$$

$$(d) \mathcal{L}^{-1} \frac{1}{(s + 1)(s + 2)^2} = e^{-t} - e^{-2t}(1 + t)$$

$$(e) \mathcal{L}^{-1} \frac{1}{s^3(s^2 - 1)} = -1 - \frac{t^2}{2} + \cosh t$$

$$(f) \mathcal{L}^{-1} \frac{s^2 + 2s + 1}{(s + 2)(s^2 + 4)} = \frac{1}{8}e^{-2t} + \frac{7}{8}\cos 2t + \frac{1}{8}\sin 2t$$

$$(g) \mathcal{L}^{-1} \frac{s^2}{(s^2 + 1)^2} = \frac{1}{2}t \cos t + \frac{1}{2}\sin t$$

7-33. Solve the differential equations given in Prob. 6-5 using the Laplace transformation method.

7-34. Solve the differential equations of Prob. 6-6 using the Laplace transformation method.

Solve the following equations using the Laplace transformation method:

$$7-35. \frac{d^2i}{dt^2} - i = 25 + e^{2t}$$

$$7-36. \frac{d^2i}{dt^2} + 4i = \sin t - \cos 2t$$

$$7-37. \frac{d^2i}{dt^2} + \frac{di}{dt} = t^2 + 2t, \quad i(0^-) = 4, \quad \frac{di}{dt}(0^-) = -2$$

7-38. Solve the differential equation given in Prob. 6-21 using the Laplace transformation method.

- 7-39. In the series RLC circuit shown, the applied voltage is $v(t) = \sin t$ for $t > 0$. For the element values specified, find $i(t)$ if the switch K is closed at $t = 0$.

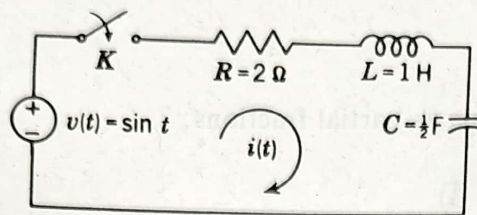


Fig. P7-39.

- 7-40. At $t = 0$, a switch is closed, connecting a voltage source $v = V \sin \omega t$ to a series RL circuit. By the method of the Laplace transformation, show that the current is given by the equation

$$i = \frac{V}{Z} \sin(\omega t - \phi) + \frac{\omega L V}{Z^2} e^{-Rt/L}$$

where

$$Z = \sqrt{R^2 + (\omega L)^2} \quad \text{and} \quad \phi = \tan^{-1} \frac{\omega L}{R}$$

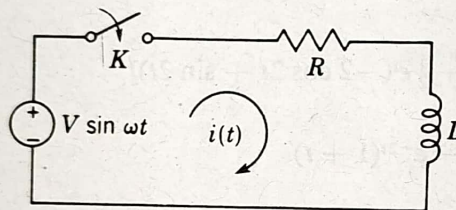


Fig. P7-40.

- 7-41. Dr. L. A. Woodbury of the University of Utah School of Medicine has made use of an electrical analog in studies of convulsions. In the network shown in the figure, the following quantities are duals: C_1 represents the volume of drug-containing fluid, R_1 is the "resistance" to the passage of the drug from the pool to the blood stream, C_2 represents the volume of the blood stream, and R_2 is equivalent to the body's excretion mechanism (kidney, etc.). The concentration of the drug dose is represented as V_0 and the voltage $v_a(t)$ at node a is analogous to the amount of drug in the blood stream. The analog network has the advantage that the elements may be readily changed and the effects studied (to say nothing of the saving of cats). Find the transform equation for $V_a(s)$ with the coefficient of the highest-order term normalized to unity.

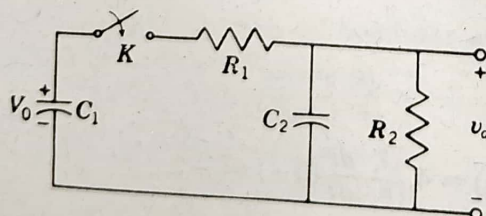


Fig. P7-41.

- 7-42. This problem is a continuation of Prob. 7-41 concerning Dr. Woodbury's analog. The following constants for the network are selected: $C_1 = 1 \mu\text{F}$, $C_2 = 8 \mu\text{F}$, $R_1 = 9 \text{ M}\Omega$, and $R_2 = 5 \text{ M}\Omega$. If $V_0 =$

100 V and the switch is closed at $t = 0$, solve for $v_A(t)$, the equivalent of the concentration of drug in the blood stream, as a function of time.

- 7-43. Find the time t_m when the concentration of drug in the blood stream for Prob. 7-42 is a maximum. (This information is desired so that a second dose may be given at that time to build up the concentration to the point where a convulsion is induced.)
- 7-44. If a second dose (the voltage equivalent having a magnitude of 100 V) is injected at $t = t_m$ as found in Prob. 7-43, what will be v_A as a function of time, and what will be the maximum v_A obtained? (Note: In giving the second dose we will assume that the total voltage is then 100 V plus the voltage on the plates at the time the addition is made.)
- 7-45. In the network shown, the switch K is closed at $t = 0$ with the network previously unenergized. For the element values shown on the diagram: (a) find $i_1(t)$, (b) find $i_2(t)$.

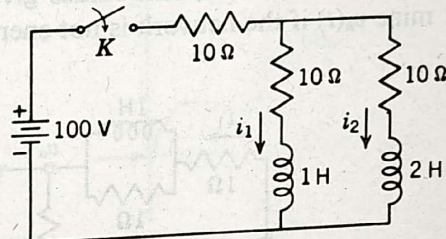


Fig. P7-45.

- 7-46. With switch K in a position a , the network shown in the figure attains equilibrium. At time $t = 0$, the switch is moved to position b . Find the voltage across R_2 as a function of time.

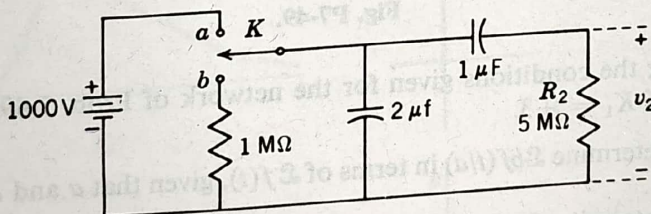


Fig. P7-46.

- 7-47. (a) Find $i_1(t)$ resulting from closing the switch at $t = 0$ with the circuit previously unenergized. The circuit constants are: $L_1 = 1$ H, $L_2 = 4$ H, $M = 2$ H, $R_1 = R_2 = 1 \Omega$, $V = 1$ V. (b) Repeat part (a) in solving for $i_2(t)$.

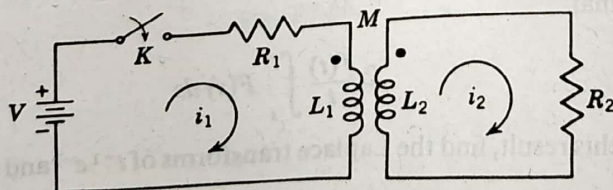


Fig. P7-47.

$$R_2 - R_2 e^{-R_1 t/L}. \quad 4-13. (b) i = (Vt/L)e^{-R_1 t/L}. \quad 4-16. (h)$$

$$i = \frac{[(a-1) \sin 2t - 2 \cos t]e^{-t} + 2e^{-at}}{(a-1)^2 + 4}$$

$$4-17. i_L = 0.6 + 0.067e^{-3.57t}. \quad 4-20. (a) R = 10^4 \Omega, (b) C = 2.5 \mu F.$$

$$4-21. (b) i = 0.01(1 - e^{-16t}) \text{ amp.} \quad 4-22. w_C/w_T = (\frac{1}{2})(1 - e^{-t/RC}).$$

Chapter 5

5-1. 0.1, -100, 100,000. 5-3. $d^2i/dt^2(0+) = -9 \times 10^5 \text{ amp/sec}^2$ 5-5. -8 V/sec², 26 V/sec³. 5-6. 0 V/sec². 5-8. 100, -10⁴, 10⁶. 5-13. (b) 0, $R_2 V/(R_1 + R_2)$, (c) V/CR_1 , 0, (d) $R_2 V/R_1 LC$. 5-15. $d^2v_1/dt^2(0+) = V_0/R$. 5-18. $1.406 \times 10^{-5} \text{ amp/sec}^2$ 5-20. (b) 66.7 V, (c) 3.33, 1.67 amps, (d) 33.3, -83,300, 5-21. $R_1 i(0+)$, 0, $R_1[(di/dt)(0+) - Ri(0+)/L]$, 0. 5-25. (a) VR_1/R_2 , (b) $(1/C) - R_1$. 5-27. (a) 0, $v(0+)/R_1 C_1$, (b) 0, 0, (c) 0, 0. 5-28. (a) $v_K(0+) = VR_3/(R_1 + R_2 + R_3)$.

Chapter 6

6-3. (f) $i = (A \cos \sqrt{1.75}t + B \sin \sqrt{1.75}t)e^{-t/2}$. (h) $i = Ae^{-2t} + Bte^{-2t}$. 6-5. (a) $i = 2e^{-t} - e^{-2t}$. (b) $i = 3e^{-2t} - 2e^{-3t}$. (f) $i = e^{-0.5t}(\cos 1.32t + 0.379 \sin 1.32t)$. (h) $i = e^{-2t} + te^{-2t}$. 6-9. $i = Ae^{-4.18t} + e^{-1.12t}(B \cos 1.06t + C \sin 1.06t)$. 6-11. $t_1 = 1/\alpha$. 6-14. $i = e^{-140t} \sin 1570t$. 6-15. $i_L = 10 \cos 316t$. 6-18. (b) $i = K_1 e^{-t} + K_2 e^{-2t} - \frac{15}{4} + (5t/2)$, (e) $v = (K_1 + t)e^{-2t} + (K_2 - 5t)e^{-3t}$. 6-23. $i = 5t^2 e^{-t}$. 6-24. $i = 10^{-4} e^{-t}(\cos t - \sin t) - 10^{-4} e^{-10^4 t}$. 6-28. $i = 10^{-3}(20.3 \cos 377t + 14.8 \sin 1414t - 0.8 \cos 1414t) \text{ amp.}$ 6-29. $s(500) = -250 \pm j968$, $s(1000) = -500 \pm j866$, $s(3000) = -2,620, -380$, $s(5000) = -4,790, -210$. 6-31. Characteristic equation: $(5 - K_1)s^2 + (6 - 2K_1)s + (2 - K_1) = 0$; the system is stable for $K_1 < 2$. 6-33. (d) $w_S = w_R + w_L = (V^2/R)[t + (L/R)(e^{-Rt/L} - 1)]$.

Chapter 7

7-9. $F_1(s) = 2/[s(s^2 + 4)]$. 7-16. $F_8(s) = s^2/[(s^2 + a^2)^2]$. 7-22. $(V/R_1)e^{-(R_1+R_2)t/L}$. 7-24. $(V/R) \cos(t/\sqrt{LC})$. 7-30. (a) $K_0 = \frac{3}{8}$, $K_2 = \frac{1}{4}$, $K_4 = \frac{3}{8}$. 7-35. $i = K_1 e^t + K_2 e^{-t} - 25 + (\frac{1}{3})e^{2t}$. 7-37. $i = (\frac{1}{3})t^3 + 2e^{-t} + 2$. 7-42. $v_a = 13(e^{-0.0215t} - e^{-0.1285t})$. 7-43. 16.7 sec. 7-44. 14 V in 29.2 sec. 7-45. (a) $3.33 + 1.21e^{-6.35t} - 4.5e^{-23.6t}$. 7-46. $1020(e^{-0.155t} - e^{-0.645t})$. 7-49. $v_a = 4 + e^{-0.75t}(-1.5 \cos 0.25t + 0.5 \sin 0.25t)$. 7-50. $i_2 = -5 - 1.34e^{-0.707t} + 16.3e^{0.707t}$.

Chapter 8

8-1. $u(t-2) - u(t-3) + u(t-7) - u(t-8)$. 8-9. $v = K_0[t^2 u(t) - 4(t-1)u(t-1) - (t-2)^2 u(t-2)]$. 8-10. Partial answer: $v = be^{(\ln(b/a)t/c} \cdot u(t) - \dots$. 8-14. (c) $i = [5 \times 10^5 t - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t) - [5 \times 10^5(t-10^{-6}) - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t-10^{-6})$. 8-17. $i(t) = [1 - e^{-(t-1)}]u(t-1) + [1 - e^{-(t-2)}]u(t-2) + [1 - e^{-(t-3)}]u(t-3) + [1 - e^{-(t-4)}]u(t-4) - 4[1 - e^{-(t-5)}]u(t-5)$. 8-19. $F_1(s) = K[(1/s^2) + (t_0/s)]$. 8-26. $V(s) = (1/s)[e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} - 4e^{-5s}]$. 8-31. $F(s) = 1/[s(1 - e^{-s})]$. 8-33. $F(s) = \{1/[(s + \frac{1}{2})(1 + e^{-(0.5+s)})]\}$. 8-34. (a) $f(t) = te^{at}$.